

Cite this: DOI: 00.0000/xxxxxxxxxx

Can PFAS adsorbents be predictively identified through machine learning? A study of PFOA-targeted screening in metal–organic frameworks[†]

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Received Date

Accepted Date

DOI: 00.0000/xxxxxxxxxx

The freshwater and drinking-water PFAS problem; Why PFOA is used as a prototype contaminant; Why database-led predictive discovery is needed; Core workflow: labelled literature set + unlabelled CoRE-MOF/CSD discovery space + simulation-informed ML refinement; Key outcome: predictive identification of chemically plausible benchmark adsorbents; General significance for PFAS sorbent discovery. Any references in the abstract should be written out in full e.g. [Surname *et al.*, *Journal Title*, 2000, **35**, 3523].

1 Introduction

1. PFAS contamination as a freshwater and drinking-water challenge
2. Why current sorbent discovery remains too slow and too empirical
3. PFOA as a chemically and regulatorily meaningful prototype among PFAS
4. Why metal–organic frameworks are attractive as tunable sorbents
5. Why the Cambridge Structural Database and CoRE-MOF subsets create a discovery opportunity
6. Why predictive design is still missing for PFAS-active metal–organic sorbents
7. Study hypothesis and paper objective

Rough draft: Per- and polyfluoroalkyl substances (PFAS) have become pervasive contaminants of freshwater and drinking-water supplies. Their widespread use in industrial processes and consumer products, coupled with the exceptional stability of their fluorinated carbon chains, has led to their persistence across aquatic

environments. Continued exposure to several PFAS has been associated with adverse health outcomes, while increasingly stringent drinking-water limits require their removal at concentrations approaching the ngL^{-1} regime. These demands present a formidable separation challenge: PFAS must be selectively scavenged from chemically complex aqueous matrices despite their low concentrations, diverse chain lengths and amphiphilic character.

Adsorption remains the most readily deployable technology for PFAS removal. Activated carbons are widely used, but their performance can be limited by slow uptake, competition from dissolved organic matter, poor affinity for shorter-chain PFAS and energy-intensive regeneration. These shortcomings have motivated the search for regenerable sorbents whose chemical environments can be tailored to recognise PFAS selectively. Metal–organic frameworks (MOFs) are particularly attractive in this regard because their pore dimensions, surface chemistry, hydrophobicity, charge distribution and metal–ligand composition can, in principle, be controlled independently. Recent studies have further shown that PFAS uptake cannot be rationalised by accessible surface area or pore volume alone. Rather, the interplay between confinement, hydrophobic association and electrostatic interactions appears to govern adsorption, highlighting the need for a multidimensional approach to sorbent design.

The chemical diversity that makes MOFs promising also makes their discovery experimentally intractable. The Cambridge Structural Database contains an extensive and continually expanding collection of experimentally reported metal–organic materials, while computation-ready subsets such as the CoRE-MOF databases provide thousands of structures suitable for systematic analysis. Only a small fraction (xx%) of this structural space has been examined for PFAS adsorption, however, and an even smaller number has been evaluated under environmentally rele-

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\footnotetext entry in the the correct place in the list.

vant conditions. Exhaustive synthesis and testing are therefore impractical, whereas high-level molecular simulations remain too computationally demanding to apply indiscriminately across the full database. Consequently, potentially exceptional adsorbents may remain hidden within a largely unexplored chemical space.

Machine learning offers a means of navigating this space, but its application to PFAS sorbent discovery is complicated by sparse and heterogeneous training data. Most reported studies describe materials that perform well, while rigorously defined inactive or poorly performing examples are scarce. Moreover, conventional geometric descriptors—including pore-limiting diameter, largest cavity diameter, accessible surface area, void fraction and framework density—cannot fully encode the chemical interactions responsible for PFAS binding. Predictive screening must therefore learn from limited labelled data while accounting for both framework geometry and chemical identity. Incorporating targeted molecular simulations into the learning cycle provides an additional route to overcome this limitation: candidates selected across the predicted performance range can be simulated, their interaction energies returned to the model, and the resulting expanded dataset used to refine subsequent predictions.

Here, we ask whether PFAS adsorbents can be prospectively identified through such an iterative, machine-learning-guided strategy. Perfluorooctanoic acid (PFOA) was selected as a prototypical target because it is a well-studied legacy C8 PFAS that combines a hydrophobic perfluoroalkyl chain with an anionic carboxylate headgroup, thereby capturing several of the competing interactions central to PFAS adsorption. We combine a curated set of MOFs with known or inferred PFOA adsorption behaviour with an unlabelled, computation-ready MOF search space. Structural descriptors calculated using Zeo++ are used to construct an initial predictive model, after which candidates spanning high-, intermediate- and low-ranking regions are subjected to molecular simulation. Iterative incorporation of the calculated PFOA–framework interaction energies enables the model to evolve from an initial sparse-label classifier towards a quantitative ranking framework.

Rather than replacing simulation or experiment, this approach uses machine learning to determine where these resource-intensive methods are most informative. In doing so, it provides both a shortlist of candidate PFOA adsorbents and a chemically interpretable assessment of the framework characteristics associated with strong binding. More broadly, the study establishes a transferable route for converting crystallographic databases into actionable discovery spaces for contaminant-selective sorbents.

2 Literature Review / Data

• Defining the chemical question and the discovery workflow

1. Literature-labelled active / inactive MOF space
2. Database-mined unlabelled discovery space
3. Relationship between Zeo++ descriptors, chemical descriptors, and adsorption behaviour

• Building the descriptor space

1. Geometric descriptors retained
2. Chemical descriptors added
3. Why topology was considered but not pursued in the final model
4. Descriptor sparsity, collinearity, and dimensionality constraints

We start with 23 PFOA active MOFs known from literature shown in Table 1. Each of these MOFs is known to be adsorption active for PFOA at various efficacy levels. However, because testing conditions across literature are unstandardised (such as MOF dosage, PFOA concentration, adsorption time, adsorption capacity, removal percentage etc.), we are unable to put a numerical score on each MOF's ability to remove PFOA. Consequently, we are only able to label these MOFs as 'PFOA active'.

We capture seven numerical features for each MOF: density, pore limiting diameter (PLD), largest cavity diameter (LCD), geometric surface area (GSA), void fraction, pore volume and molecular weight per unit cell, also shown in Table 1. Large enough PLD and LCD are expected to allow PFOA adsorption, without being so large so as to not hold tightly to the PFOA molecule. These features are meant to capture the necessary (but not sufficient) condition for adsorption. GSA, void fraction and pore volume are expected to capture the capacity potential of adsorption, on the other hand. We also information available regarding the metal active node of the MOF and its density and molecular weight per unit cell for potential use in machine learning modelling.

3 Methodology

The ML–simulation based workflow starts from the 23 PFOA active MOFs in existing literature and is formed of three major steps:

1. Baseline model (OCSVM) + binary simulation **1 iteration**
2. SVM + energy simulation **1 iteration**
3. SVR + energy simulation **iterations**

We note novelty of architecture and application using existing machine learning modelling approaches.

3.1 Baseline Single Class Labelling model (OCSVM)

The discovery of new PFAS-removing MOFs presents a challenge for conventional supervised learning approaches, as reliable negative examples are inherently difficult to define. While a subset of experimentally validated MOFs exhibiting strong PFAS adsorption performance can be identified from literature, the vast majority of available MOF structures lack corresponding adsorption measurements and cannot be confidently assigned as inactive. This creates a positive-unlabelled learning problem rather than a conventional binary classification task.

To address this challenge, we employ a One-Class Support Vector Machine (OC-SVM) approach, which is specifically designed to learn the characteristic feature space occupied by a single class. Unlike traditional support vector machines, which construct a

#	Name	Metal node	Density (g/cm ³)	PLD(Å)	LCD (Å)	GSA (m ² /g)	Void fraction	Pore volume (cm ³ /g)	Molecular weight of unit cell(Å ²)
1	UiO-66	Zr(IV)	1.23	3.82	8.70	2,921	0.59	0.48	6,640
2	UiO-66-NH ₂	Zr(IV)	2.02	1.28	7.33	2,383	0.42	0.21	10,960
3	UiO-66-F ₄	Zr(IV)	1.64	1.08	2.72	4,812	0.28	0.17	1,381
4	UiO-67	Zr(IV)	0.72	6.73	12.97	3,475	0.74	1.02	8,467
5	UiO-67-F ₂	Zr(IV)	0.86	6.64	12.83	3,059	0.72	0.84	10,194
6	MOF-808	Zr(IV)	1.28	10.09	15.64	2,178	0.63	0.49	33,373
7	PCN-999	Zr(IV)	0.59	18.48	19.44	3,430	0.79	1.34	19,228
8	PCN-1002	Zr(IV)	1.27	2.06	7.42	3,526	0.52	0.41	7,116
9	DUT-67	Zr(IV)	1.03	6.63	14.51	2,708	0.70	0.68	36,239
10	NU-1000	Zr(IV)	0.49	28.35	28.81	3,605	0.82	1.68	6,482
11	MIL-53(Al)-as	Al(III)	1.00	5.95	6.52	3,991	0.59	0.59	13,254
12	MIL-96(Al)	Al(III)	1.15	3.63	8.33	3,579	0.56	0.49	3,837
13	MIL-100(Al)	Al(III)	0.68	8.50	27.81	3,593	0.74	1.09	150,561
14	MIL-101(Cr)	Cr(III)	0.46	13.84	34.15	3,805	0.83	1.80	193,355
15	MIL-101(Cr)-NH ₂	Cr(III)	1.87	1.16	2.33	4,245	0.36	0.19	4,081
16	MIL-100 (Fe)	Fe(III)	0.76	8.85	27.76	3,271	0.74	0.97	180,331
17	MIL-101 (Fe)	Fe(III)	0.46	13.84	34.15	3,805	0.83	1.80	193,355
18	ZIF-7	Zn(II)	1.44	3.13	4.35	3,930	0.44	0.31	2,076
19	ZIF-8	Zn(II)	0.92	3.58	11.98	4,996	0.64	0.69	2,707
20	ZIF-L	Zn(II)	1.50	1.07	2.36	5,397	0.35	0.23	7,344
21	HKUST-1	Cu(II)	0.95	5.21	11.12	3,707	0.69	0.72	10,446
22	Mn-BTC	Mn(II)	1.87	0.62	1.50	4,112	0.19	0.10	2,986
23	Ce-BTC	Ce(III)	1.28	6.40	6.62	3,273	0.57	0.44	1,184

Table 1 Initial PFOA active list

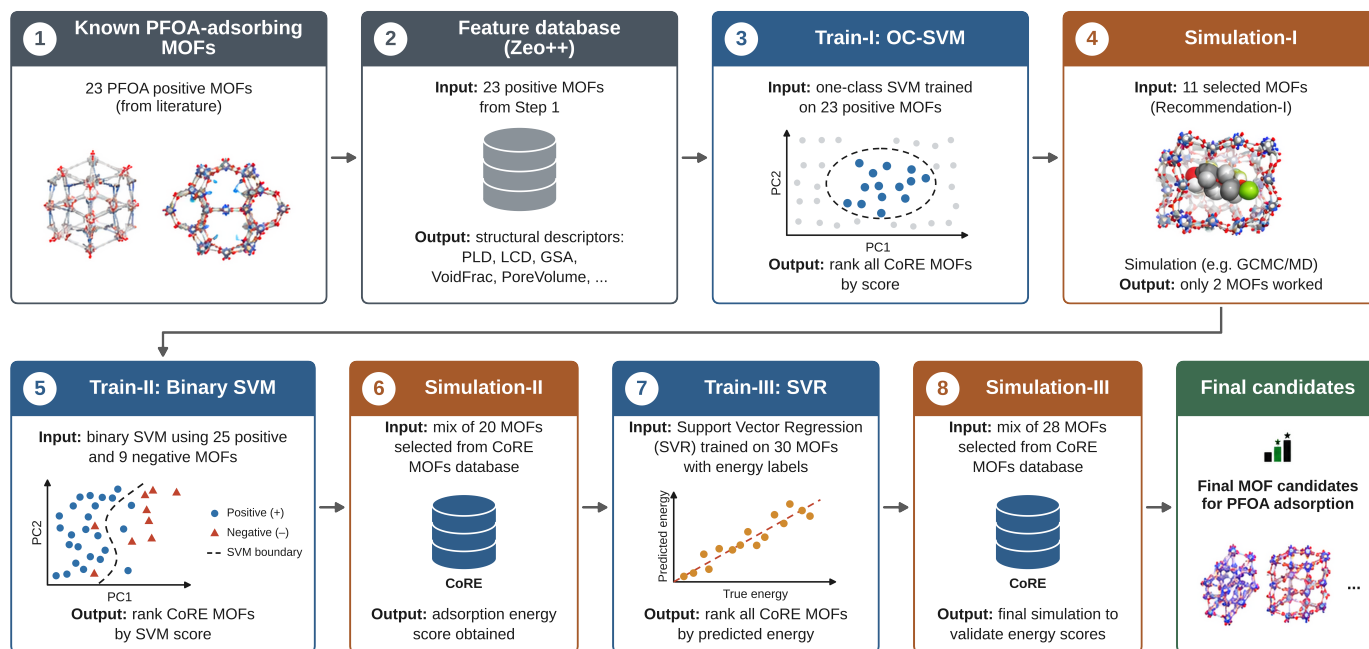


Fig. 1 Overall workflow.

maximum-margin classifier separating positive and negative samples, OC-SVM identifies a boundary enclosing the region of feature space associated with known positive examples. Although commonly applied for anomaly detection, including fraud identification and cybersecurity applications, the same principle can be adapted for materials discovery by defining successful PFAS-removing MOFs as the target class. The resulting model learns

the structural and chemical characteristics associated with effective PFAS adsorption and enables the screening of unexplored MOFs within the CoRE MOF database prior to computationally expensive molecular simulations.

The descriptor vector of each experimentally reported PFAS-

active MOF is represented as:

$$\mathbf{x}_i \in \mathbb{R}^d, \quad i = 1, \dots, n, \quad (1)$$

where d represents the number of physicochemical descriptors used to define the MOF feature space.

The OC-SVM attempts to separate the mapped data from the origin by finding a hyperplane:

$$\mathbf{w}^T \phi(\mathbf{x}) - \rho = 0, \quad (2)$$

where:

- $\phi(\mathbf{x})$ is called the kernel. It allows for the features to be transformed to a higher dimension space thus allowing non-linear separating regions (instead of hyperplanes) to enclose (rather than separate) the data. We use the RBF kernel.
- $\mathbf{w}$ are the coefficients on the now linear-in-kernel (not features) hyperplane. These can be thought of as the gradient or orienting parameters that can be manipulated to determine the separation of the hyperplane from the origin.
- ρ is the constant parameter that functions similarly in correctly positioning the hyperplane.

While the details of solving the optimisation problem that determine the best set of values for the orienting and constant parameters are not needed, we none the less briefly present the optimisation problem in order to motivate the hyper-parameters we must control. The OC-SVM is obtained by solving the following optimisation problem:

$$\begin{aligned} \min_{\mathbf{w}, \rho, \xi} \quad & \frac{1}{2} \|\mathbf{w}\|^2 + \frac{1}{vn} \sum_{i=1}^n \xi_i - \rho \\ \text{subject to} \quad & \mathbf{w}^T \phi(\mathbf{x}_i) \geq \rho - \xi_i, \quad i = 1, \dots, n, \\ & \xi_i \geq 0, \quad i = 1, \dots, n. \end{aligned} \quad (3)$$

The minimisation across the two parameters w and ρ fits the smallest boundary encompassing the data, however the introduction of an error term ε allows fitting a tighter boundary, with the hyper-parameter ν controlling the degree to which errors may be accepted. Figure 2 shows the OC-SVM results based on hyper-parameter value of $\nu = 15\%$ (values between 5% to 20% showed similar results). The 23 PFOA active MOFs together define a region in the feature space that is visualised using a 2-PCA plot. We see two basins of attraction one on each side. Zif-7 and MIL-101(Cr)-NH₂ are the highest scored MOFs i.e the most central to the region. Both these MOFs are found in the left basin of the feature plot. On the other hand, five MOFs are most further apart from the OC-SVM region and are shown in blue: Zif-8 (top), MIL-101(Cr) and MIL-101(Fe) (top right - overlapping), MOF-808 (bottom right) and PCN-999 (appears central - boundary between the two basins of attraction). Finally, we remark the some MOFs may appear further away from the centre but this is a artefact of the 2-PCA projection working hard to capture information from higher dimensional space. (Table of actual training results can be put in SI).

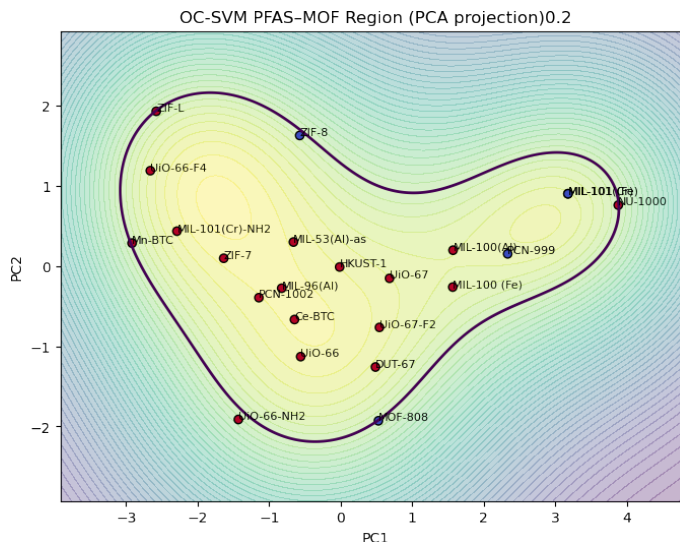


Fig. 2 OCSVM results classifying the original 23 MOFs*.

Once we have trained the OCSVM model, we are able to pass the features of any generic MOF and obtain a centrality score for the MOF. A high positive score indicates MOFs central to the region defined by the model, negative scores mean MOFs outside the region while a score of zero defines MOFs that lie on the boundary of the region and can be described as ‘indeterminate’. More extreme positive/ negative scores are commensurate of greater/ lesser ‘qualitative’ nearness to the central region defined by the model.

Our next step becomes the validation of the OCSVM model via simulations, if the model is well defined highly positive scored MOFs will return positive simulation while negatively scored ones will return negative simulation results. However, a point to note here is while simulations represent lower effort than physical experimentation, they none the less are time consuming.

The simulation methodology we used... (Souza to provide details of the simulation)

Table 2 represent a choice of the six highest scored MOFs from the CoRE MOFs database along with the three worst and two indeterminate MOFs, the six highest scored ones are predicted to be PFOA ‘Active’, the three worst scored MOFs are predicted to be In-Active while the borderline ones are assigned the ‘Indeterminate’ label. The last column represents the results of carrying out simulations on these MOFs. As can be seen ‘Active’ predictions were all incorrect, while the ‘Indeterminate’ ones came back PFOA active. Therefore an improvement in the learned boundaries of the model was needed. However, there was still some value to be found in the the model as it was able to correctly predict the simulation outcome for the ‘InActive’ MOFs.

While these results are mixed, they give the necessary information to further refine the machine learning approach.

* *Reference Table 1*

Name	OCSVM ($v = 0.15$)	Density	PLD	LCD	GSA	Void Fraction	Pore Volume	Molecular Weight (g/mol)	Stage 1 OCSVM Prediction	Stage 1 Simulation Result
2008[Co][sql]2[ASR]1	0.15	1.52	2.43	4.78	4,256	0.40	0.26	2,113	Active	NotGood
2018[Co][flu]3[ASR]1	0.15	1.35	2.80	4.45	4,348	0.43	0.32	917	Active	NotGood
2022[Co][gra]3[ASR]1	0.15	1.30	2.18	5.07	4,527	0.45	0.34	3,940	Active	NotGood
2015[Pr][hcb]2[ASR]1	0.15	1.51	2.33	4.48	4,281	0.41	0.27	3,480	Active	NotGood
2023[Zn][nan]3[ASR]13	0.15	1.51	2.31	4.47	4,462	0.41	0.27	2,909	Active	NotGood
2024[Cu][pcu]3[ASR]2	0.15	1.42	2.33	4.03	4,418	0.41	0.29	5,405	Active	NotGood
2021[Eu][fcu]3[ASR]2	0.00	0.79	7.21	14.75	3,040	0.76	0.96	2,505	InDet	Active
2022[Ni][kgd]2[ASR]1	- 0.00	0.83	3.23	9.75	5,091	0.61	0.73	946	InDet	Active
2015[Zr][ftw]3[ASR]3	- 0.65	0.29	10.50	25.26	4,625	0.87	3.00	4,241	InActive	InActive
2016[Mg][stp]3[ASR]2	- 0.67	0.28	26.59	28.06	4,802	0.88	3.08	2,538	InActive	InActive
2020[Fe][hcb]2[ASR]2	- 0.78	0.18	15.58	22.95	4,288	0.93	5.31	792	InActive	InActive

Table 2 OCSVM prediction and simulation results for 11 selected COREMOFs

3.2 Two-Class Classification Model (SVM)

In the next step, we are able to append the 11 CoRE MOFs from Table 2 to the original 23 PFOA-active MOFs to obtain a dataset of 34 MOFs, comprising 25 positive and 9 negative samples. The labels of the additional MOFs were assigned from the molecular simulation results, with a positive label corresponding to a PFOA-active MOF and a negative label corresponding to a PFOA-inactive and Indeterminate MOFs. This additional simulation-derived information transforms the initial positive-only learning problem into a conventional two-class classification problem.

A Support Vector Machine (SVM) was therefore employed in step 2, to learn a decision boundary between the active and inactive MOFs. Given a set of labelled MOFs represented by descriptor vectors $\mathbf{x}_i$ and corresponding class labels $y_i = -1$ for inactive and $y_i = +1$ for active, the separating hyperplane for a 2-class learning problem can be expressed as

$$\mathbf{w}^T \phi(\mathbf{x}) + b = 0. \quad (4)$$

As before; $\mathbf{w}$ defines the orientation of the hyperplane, b is the intercept, and $\phi(\mathbf{x})$ maps the input descriptor vector into the feature space via the kernel. However the optimisation problem that defines the optimal hyperplane looks much different this time and is obtained by maximising the margin i.e. the separation, between the two classes. For a linearly separable in kernel dataset, this can be formulated as the minimisation problem

$$\begin{aligned} \min_{\mathbf{w}, b} \quad & \frac{1}{2} \|\mathbf{w}\|^2 \\ \text{subject to} \quad & y_i (\mathbf{w}^T \phi(\mathbf{x}_i) + b) \geq 1, \quad i = 1, \dots, n. \end{aligned} \quad (5)$$

In practice, perfect separation of the active and inactive MOFs is not expected, particularly given the limited number of simulation-labelled samples. We therefore use the soft-margin formulation (6), which introduces slack variables ξ_i to permit individual observations to violate the margin:

$$\begin{aligned} \min_{\mathbf{w}, b, \xi} \quad & \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \xi_i \\ \text{subject to} \quad & y_i (\mathbf{w}^T \phi(\mathbf{x}_i) + b) \geq 1 - \xi_i, \\ & \xi_i \geq 0, \quad i = 1, \dots, n. \end{aligned} \quad (6)$$

Here C controls the trade-off between maximising the classification margin and penalising misclassified or margin-violating observations. A large value of C places greater emphasis on correctly classifying the training data, whereas a smaller value permits greater tolerance of classification errors in favour of a wider margin.

For nonlinear separation of the MOF descriptor space, the mapping $\phi(\mathbf{x})$ can be avoided explicitly through the use of a kernel function,

$$K(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j). \quad (7)$$

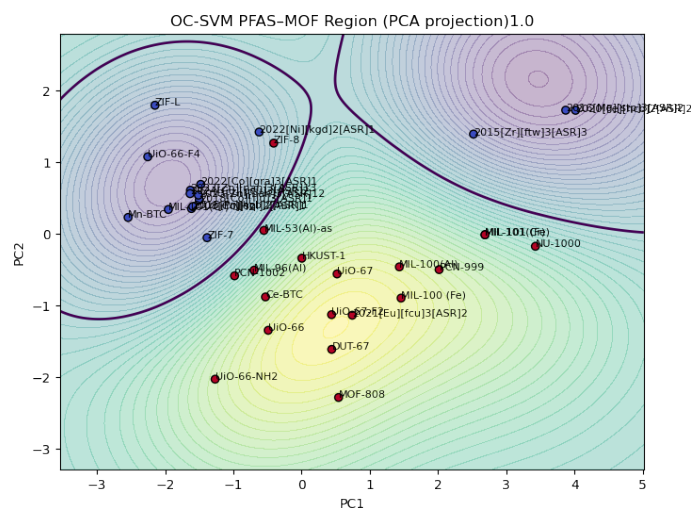
The resulting classification function can then be written in its kernelised form as

$$f(\mathbf{x}) = \sum_{i=1}^n \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b, \quad (8)$$

where α_i are the coefficients obtained during optimisation and the observations with non-zero α_i constitute the support vectors. The predicted class is determined from the sign of the decision function,

$$\hat{y} = \text{sign}[f(\mathbf{x})]. \quad (9)$$

The SVM was subsequently trained using the combined set of literature-derived and simulation-labelled MOFs and applied to the remaining MOFs within the CoRE database. In contrast to the OC-SVM, which identified MOFs according to their similarity to the descriptor space of known PFOA-active materials, the two-class SVM is able to explicitly distinguish between descriptor characteristics associated with simulated PFOA-active and PFOA-inactive MOFs. Similar to the OC-SVM, higher positive scores indicate stronger class membership while increasing negative scores



indicate a stronger membership to the class of non-PFAS absorbent MOFs. Figure 3 shows the results of the trained model, we see a recalibration of the model boundaries, ascertainable from the fact both the previously most central MOFs are now in the out-region, Zif-7 (bottom right of the left circular out-region) and MIL-101(Cr)-NH₂ (bottom right central to the left circular out-region).

As before, we must validate the SVM results in a further iteration of simulation. Table 3 shows the 20 MOFs we select for the next batch of simulations. We again choose a mix of 12 high scoring, 4 indeterminate and 4 low scoring MOFs, this time adding the CORE MOFs subsets ION and FSR in addition to the previous ASR MOFs. The simulation results are captured in the column ‘Normalised EnergyScore (NES)’, we find the 12 MOFs scored highly by the SVM model to have high NES, while the 4 highly negative SVM scored MOFs do not return an energy measure from the simulation signifying their inability to adsorb PFAS. Of the 4 Indeterminate MOFs, 2 are as expected while the other 2 return outlier values of high energy scores. We use this information to further develop the ML model.

3.3 Support Vector Regression

More negative suggests the PFOA binding energy is higher.

The conclusions section should come in this section at the end of the article, before the Author contributions statement and/or Conflicts of interest statement.

Conflicts of interest

Acknowledgements

Name	SVM ($C = 1.0$)	Normalised EnergyScore	PLD	LCD	GSA	Void Fraction	Pore Volume	UnitCell_MW (g/mol)
2022[Zr][lvt]3[ASR]1	1.44	0.38	6.93	9.83	2,837	0.67	0.63	5,783
2021[Hf][sqc]3[FSR]1	1.41	0.27	5.25	9.20	2,701	0.67	0.55	4,928
2021[Eu][fcu]3[FSR]1	1.41	0.35	7.21	13.71	3,103	0.75	0.91	2,601
2016[Tb][fcu]3[FSR]1	1.41	0.29	6.95	13.38	3,091	0.74	0.93	2,531
2021[Th][fcu]3[FSR]1	1.40	0.27	7.12	13.68	2,785	0.74	0.80	1,665
2024[Tb][tfz]3[FSR]1	1.40	0.29	8.87	9.74	2,788	0.71	0.70	4,693
2020[Zn][nan]3[ASR]16	1.35	0.35	11.01	11.72	2,699	0.67	0.55	975
2023[La][pts]3[ASR]1	1.33	0.55	7.50	8.09	3,002	0.62	0.51	870
2017[Co][nan]3[ASR]5	1.22	0.23	5.76	7.70	2,945	0.56	0.41	1,104
2018[ZnMo][qtz]3[ION]2	1.18	0.27	3.78	6.05	2,445	0.50	0.24	3,987
2018[ZnMo][qtz]3[ION]1	1.18	0.16	3.78	6.05	2,445	0.50	0.24	3,987
2018[Ce][csq]3[ASR]1	1.10	0.33	17.74	23.52	3,477	0.81	1.44	5,453
2016[La][sql]2[ION]1	0.07	-	2.63	3.25	3,303	0.34	0.16	1,015
2020[Pr][nan]3[FSR]1	0.00	0.78	4.39	7.71	3,726	0.48	0.31	4,371
2020[Cu][nbo]3[FSR]1	- 0.00	0.31	10.01	10.63	4,825	0.58	0.59	1,380
2020[ReAg][pts]3[ION]1	- 0.02	-	2.29	4.49	3,524	0.44	0.23	813
2019[Cu][sql]2[ION]1	- 1.09	-	2.94	4.75	4,550	0.40	0.28	1,360
2017[In][kgd]2[ION]1	- 1.10	-	3.12	4.88	4,467	0.38	0.26	4,874
2016[Zn][nan]2[FSR]1	- 1.12	-	2.28	3.77	4,473	0.39	0.25	1,399
2017[Ni][hms]3[FSR]1	- 1.14	-	2.34	3.06	4,474	0.37	0.25	2,532

Table 3 SVM results and structural properties for 20 selected MOFs